

Federated Learning TP3

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Course: AI for Connected Industries

Module: Distributed and Federated Learning

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Abstract—This report presents the implementation and evaluation of TP3 from the Federated Learning course laboratory. A federated learning pipeline was developed using the Flower framework, with Fashion-MNIST as the benchmark dataset. TP2 explored the effects of data heterogeneity, while TP3 focuses on the impact of data and model poisoning attacks. We evaluate two robust aggregation strategies, FedMedian and Krum, to analyze their effectiveness under varying malicious client ratios and data heterogeneity levels.

Index Terms—Federated Learning, model and data poison, fedmedian, krum

I. INTRODUCTION

Federated learning is increasingly deployed in safety-critical applications such as healthcare, mobile personalization, and autonomous systems. However, its decentralized nature exposes it to poisoning attacks, where malicious clients manipulate local training to corrupt the global model. Label-flipping data poisoning and scaled model poisoning are evaluated against FedMedian and Krum under varying malicious ratios and heterogeneity.

II. REPORT STRUCTURE

The end to end orchestration and experimentation of federated learning is hosted on GitHub for replicability. (Git Repo: dfl-tp-workspace). Report covers the results from different experiments on flower where various hyperparameter values are tested in combination. The plots from every tp experiments are recorded in a single pdf added as annexure (annexure-output-plots.pdf).

III. BACKGROUND

1) *Data and Model Poisoning*: Federated learning is vulnerable to malicious clients that attempt to corrupt the global model. Attacks exploit the fact that the server in standard FedAvg has no mechanism to verify the integrity of client updates.

Data poisoning occurs when a client trains on manipulated local data, for example by flipping class labels, and sends the resulting poisoned weight update to the server. The update appears structurally normal but encodes incorrect class boundaries.

Model poisoning is more direct where a malicious client computes and sends arbitrary weight updates, regardless of its local data, with the explicit goal of degrading global accuracy or causing the model to misclassify specific targets.

For purposes of experiments gradient direction reversal and scaling attack (scale factor s) is utilized.

2) *FedMedian*: FedMedian replaces the weighted mean aggregation of FedAvg with a coordinate-wise median for each model parameter, the server takes the median value across all client updates rather than their average. Because the median is statistically robust to outliers, a small number of malicious clients submitting extreme weight values cannot shift the aggregate arbitrarily. FedMedian therefore tolerates up to $\lfloor (n-1)/2 \rfloor$ clients without corrupting the aggregated model, at the cost of slower convergence.

3) *Krum*: Krum selects a single client update to use as the global model for each round, rather than averaging all updates. For each client Krum computes the sum of squared Euclidean distances from the client i 's update to its $n-f-2$ nearest neighbors, where f is the assumed number of malicious clients. The client with the smallest such score that is, the update most similar to the majority of other updates, is selected. Malicious updates that are far from the benign cluster receive high scores and are rejected. Krum provides a Byzantine-fault-tolerance guarantee when $n \geq 2f + 3$, but discards useful information from benign clients not selected, which reduces statistical efficiency relative to FedAvg in the absence of attacks.

IV. EXPERIMENTS

TABLE I: Default Setup: Fixed and varied parameters for all experiments.

Parameter	Value
Dataset	FashionMNIST
Model	CNN (2 conv layers + 2 FC layers)
Learning rate	0.01
Clients n	50
Fraction train	1
alpha	10
Local epochs E	3
Batch size	64
Communication rounds	20
Data poisoning	label-flipping (0 $\rightarrow$ 9)
Model poisoning	$\delta w_{poison} = -s \cdot \delta w_{benign}$
malicious client	{0, 25%, 50%}
Recovery Strategies	{Fedmedian, Krum}

A. Impact of data and model poisoning at different levels of malicious clients

1) *Expected Behavior*: In the absence of adversarial attacks, fedavg produces a centralized model whose accuracy improves

consistently as more rounds are aggregated. Under poisoning attacks, the expected behavior differs: Data poisoning should degrade global evaluation accuracy while leaving training loss consistently reduces, due to poisoned local data creating misleading gradients. **Model poisoning** is expected to cause severe instability or collapse of global accuracy, since malicious clients directly manipulate model updates.

2) Observation and Deviations:

- Data poison attack with malicious clients = 25%, it has limited impact, as honest client gradients dominate the global model updates. This matches the expected signature of data poisoning: high training accuracy but suppressed global generalization. (refer Figure 5)
- Conversely, Higher malicious (50%) ratios amplify the effect of distortion, opposite gradients keeps the gradients oscillating without converging. (refer Figure 1)
- Model poison is more devastating on the learning. With a scaling factor of 2, the learning is drastically slowed and as malicious clients increases to 50% fedavg fails to converge. The inconsistent spikes in federated accuracy reflect poisoned gradients overpowering benign updates.
- The degradation is non-linear: it accelerates as malicious ratio increases.

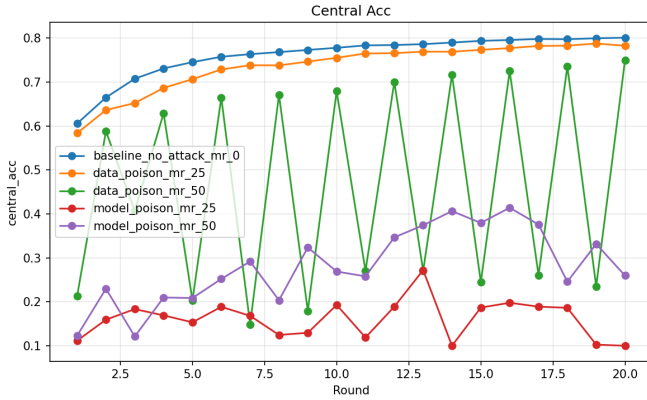


Fig. 1: Central Test Accuracy data and model poisoning attacks

B. Effect of Fedmedian and krum as recovery strategies

1) *Expected Behavior:* Robust aggregation rules such as Fed-Median and Krum are designed to mitigate the influence of poisoned clients. Under **no attack**, all aggregators should converge similarly to FedAvg. Under **data or model poisoning**, FedMedian should provide moderate robustness by reducing the impact of outlier updates. Krum should theoretically select the closest benign updates and reject poisoned ones, especially when the malicious ratio is below 50%. As the malicious ratio increases (25% $\rightarrow$ 50%), both methods are expected to degrade.

2) Observation and Deviations:

- Under no attack conditions, fedmedian slightly under performs relative to baseline. While krum is more vulnerable

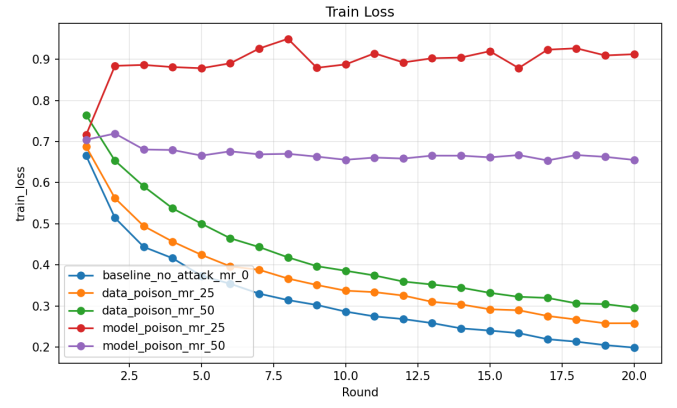


Fig. 2: Training loss for combination of alpha and malicious client volume

and tend to be less stable as single client updates are utilized. Fedavg largely starts failing at 50% malicious clients.

- **Data Poisoning:** The Fedmedian strategy largely suppresses attacks when fewer malicious clients are involved, as the median is dominated by honest client updates. As malicious clients grow beyond the tolerance levels of $\lfloor (n-1)/2 \rfloor$, it fails to converge, although remains stable. This is in line with theoretical expectations (refer Figure 3).
- Similarly krum also recovers to an extent for malicious clients = 25% with data poisoning, but under-performs compared to fedmedian. As krum relies on distance metrics, it degrades when malicious clients = 50% where all clients form coherent clusters without large distinction. (refer Figure 3)
- **Model poisoning:** With fewer malicious clients, both fedmedian and krum recovers well. As model poisoning generates outliers, median based and distance based methods perform well.
- Malicious clients form distinct clusters in model poison attacks as gradients are largely different. As a result krum effectively counters the attack even with large number of malicious clients. On the other hand, fedmedian fails even with model poisoning as malicious clients grow near to 50%.
- Although model poisoning is far more devastating unchecked, it is relatively reversible with strategies such as krum, unlike data poison attacks which can inherently affect gradients.

C. Impact of heterogeneity at different levels of malicious clients

1) *Expected Behavior:* The parameter α controls the degree of data heterogeneity across clients. For **small** α (e.g., 0.01), client data is highly non-IID. Robust aggregation methods (FedMedian, Krum) should help stabilize training under poisoning. For **large** α (e.g., 10), the data is more IID. Poisoning

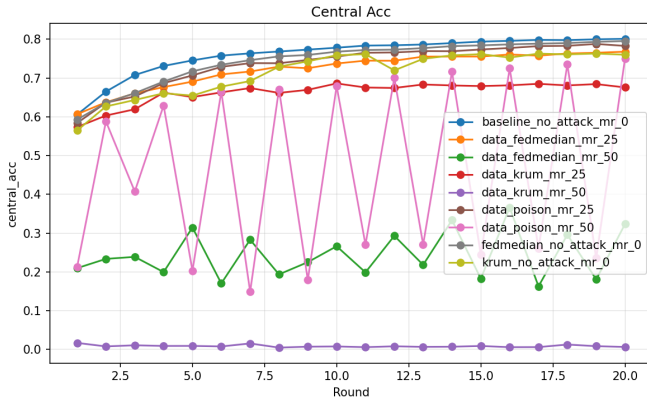


Fig. 3: Central Test Accuracy for fedmedian and krum on data poison attacks

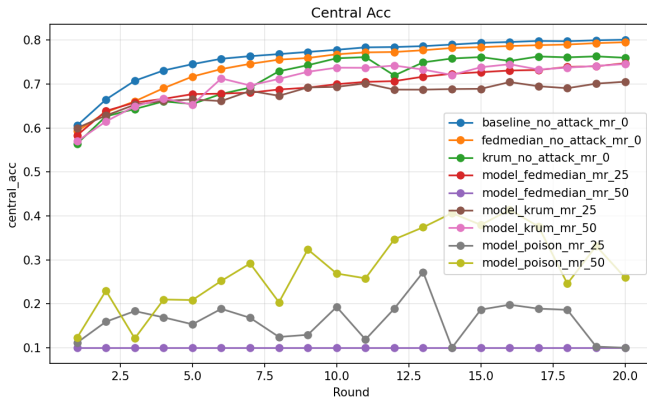


Fig. 4: Central Test Accuracy for fedmedian and krum on model poison attacks

attacks are expected to be easier to detect and mitigate, and robust aggregators tend to perform closer to the clean baseline.

2) Observation and Deviations:

- Model poisoning completely distorts learning for both non-IID and IID data. (refer Figure 5)
- As expected, higher α yields smoother convergence for baseline without attacks. Also, the learning with counteractive strategies such as fedmedian and krum also stays closer to baseline.
- Fedmedian, although provides better performance to counter model poisoning for IID data, it fails to recover for heterogeneous clients at $\alpha = 0.01$. Non-IID data makes benign gradients inconsistent, allowing poisoned updates to dominate.
- Similar to fedmedian, krum also fails to counter model poisoning with non-IID conditions. As gradient updates are dependent on single client update and few classes dominate in non-IID data state, no updates are holistic to converge at global level. Multi-krum offers a trade-off to improve krum performance.
- Both krum and fedmedian work best when malicious

clients are under 50%. As a result, FedMedian or Krum rejects honest clients with outlier behavior when they are minority outliers as the assumption honest clients geometrically outweigh malicious clients collapses.

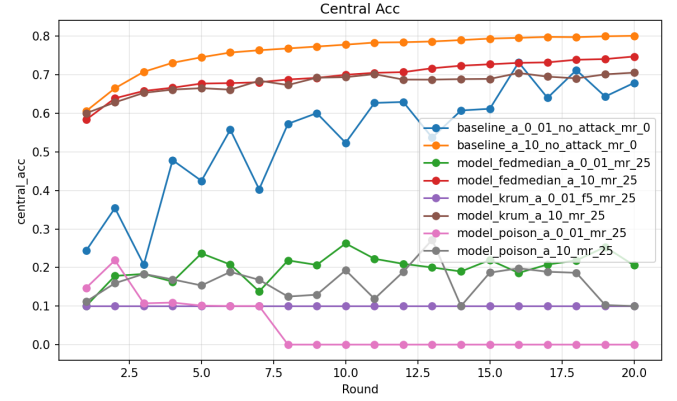


Fig. 5: Central Test Accuracy for combination of alpha and malicious client volume

V. CONCLUSION

Data poisoning is a persistent but moderate threat under FedAvg. At 25% malicious ratio it slows convergence without preventing learning. At 50% clients create opposing gradients that keeps the model oscillating.

Model poisoning is catastrophically more effective. At 25% malicious ratio with scale=2, FedAvg is completely destroyed after few rounds. The geometric negative scaling repeatedly undoes the previous round's learning, creating a permanent equilibrium at the initialization point.

FedMedian provides strong protection against both attacks when fewer clients are malicious. However, FedMedian fails at exactly 50% poisoning and at the boundary, the median falls in the corrupted zone. FedMedian is the practical defense as it requires no knowledge of malicious client volume.

Krum protects against model poisoning when the malicious count is correctly estimated. Its weakness is that it selects a single client update discarding all other information and can lead to slower convergence. Against data poisoning, Krum fails as the malicious clients grow because poisoned clients form a coherent cluster that Krum cannot distinguish from the honest cluster. Krum is appropriate only when the attack type is magnitude-based.

Heterogeneity Under high heterogeneity ($\alpha=0.01$), honest clients have distinct gradients and do not form coherent clusters. This distorts them as malicious clients to both FedMedian and Krum undermining recovery. The defense that protects against attacks under IID conditions progressively misidentifies honest non-IID clients as suspicious under increasing heterogeneity. Simultaneously handling both high heterogeneity and high malicious ratio is a tradeoff that requires robust design.